# Pricing the Privacy–Utility Trade-off

**How to build the curve, compare techniques and policies, and put the choice in front of decision makers**

Study notes

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Differential privacy (DP) buys a formal guarantee by adding noise, and the noise costs accuracy. Engineers usually show that cost as a curve of accuracy against the privacy parameter $\varepsilon$, which means little to the people who have to approve a release. This note shows how to turn both axes into units a business already uses, how to build a privacy–utility Pareto frontier that compares different DP techniques fairly, how organisational privacy budgets enter the choice, and how to present the decision as a short menu of options. It draws on research papers, government guidance and published deployments, which are listed at the end.

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# 1 Why the curve has to be priced

Every differentially private system has a dial. Turning it one way adds noise and strengthens the guarantee. Turning it the other way removes noise and makes the outputs more accurate. The dial is usually labelled with $\varepsilon$, the privacy-loss parameter, where a smaller value means a stronger guarantee. An engineer can sweep that dial and draw a curve of accuracy against $\varepsilon$, and the curve is the honest technical description of the trade-off.

That curve is rarely enough to make a decision, though. A product owner cannot weigh "$\varepsilon = 2$" against "0.91 AUC", because neither number is in a unit they manage. The US Census Bureau put the problem plainly when it prepared the 2020 census. Its staff described accuracy against privacy loss as a fundamental trade-off. Choosing a point on it, they said, is a social choice that belongs to policy makers, within the limits of what the technology can do (Census Bureau, 2019). Inside a company the same holds. The choice belongs to whoever owns the risk and the revenue, and the engineer's job is to give them a curve they can read.

So this note has two halves. The first half (Section 2 and Section 3) translates the axes. Utility becomes money or a key performance indicator, and privacy becomes the odds of a concrete harm. The second half (Section 4 to Section 7) builds the curve properly, as a Pareto frontier over many techniques and settings, and then fits it to the budgets that an organisation's privacy policy sets.

# 2 Putting both axes in business units

## 2.1 The utility axis

Model accuracy and root-mean-square error are the engineer's measures of utility, but the business cares about what the lost accuracy does downstream. The most direct translation runs the private and the non-private model through the evaluation the business already trusts. For a fraud model that is the fraud caught at a fixed review capacity. For a recommender it is conversion in an offline replay or an A/B test, and for a demand forecast it is the cost of stock-outs and overstock. A statement such as "moving from no DP to $\varepsilon = 4$ costs 0.6% of the fraud we catch, about €400,000 a year" is something a budget holder can weigh.

Published statistics need a different translation, because nobody consumes a dashboard's error directly. What matters is whether the noise ever changes a decision. Take a count released with the Laplace mechanism, which adds noise whose typical size depends on two things. The sensitivity $\Delta$ is the most that one person can change the count, which is 1 for a plain count. The noise scale is then $\Delta/\varepsilon$, and that is also the average size of the error:

$$\mathbb{E}\,|\text{noise}| = \frac{\Delta}{\varepsilon}.$$

Take a count of 2,000 customers released at $\varepsilon = 0.5$. Its typical error is 2 customers, or 0.1%. If the dashboard is read to the nearest hundred, that error never changes a decision, and the utility cost is zero. The useful metric here is therefore a *decision error rate*. It counts how often the noisy release would lead somewhere the true value would not. Examples are ranking two regions in the wrong order, crossing a reorder threshold, or flagging an anomaly that is not there. That rate can be estimated by simulation on historical data before anything is released.

The same example shows where the real cost usually lies. The error of 2 customers is fixed by $\Delta$ and $\varepsilon$, whatever the size of the count, so it weighs far more on a small group. On a segment of 40 customers it is 5%. In general the relative error of a count or a mean over $n$ people falls like

$$\frac{\text{error}}{\text{true value}} \approx \frac{\Delta}{n\,\varepsilon},$$

so a dataset twice the size reaches the same accuracy at half the $\varepsilon$. Small regions, rare products and minority populations therefore pay most of the cost of privacy. Microsoft made this visible when it released US broadband coverage by ZIP code with DP. The noise mattered most in ZIP codes with few households, so Microsoft published an error range for each ZIP code alongside the data. It also showed that publishing the ranges cost no extra privacy (Microsoft Research, 2021). A utility report should do the same and give the cost for each segment the business cares about, not only the average.

The last part of the utility axis is the cost of buying accuracy back. At a fixed $\varepsilon$, utility can often be recovered with more data, larger training batches, more compute or a model pre-trained on public data. DeepMind's work on image classification is the clearest example. A pre-trained model fine-tuned with DP reached 86.7% top-1 accuracy on ImageNet at $\varepsilon = 8$, about four points below the non-private state of the art (De et al., 2022). Tramèr and Boneh showed the reverse. Without good features or much more data, deep networks trained with DP lost to linear models on handcrafted features (Tramèr & Boneh, 2021). Putting a price on the buy-back turns "we lose two points of accuracy" into "we spend €50,000 more on compute and lose nothing", which is often an easier decision.

### 2.2 The privacy axis

On its own, $\varepsilon$ is a number without units or context, and that makes it hard to explain (CDT, 2023). A user study by Nanayakkara and colleagues tested several ways of explaining it to people deciding whether to share their data. Explanations given as odds worked better than the alternatives. People who were told the value of $\varepsilon$ were also more willing to share, and less so as the guarantee weakened (Nanayakkara et al., 2023). The privacy axis should therefore be stated as odds.

The cleanest odds statement concerns an attacker who wants to know whether one person, Alice, is in the data. Suppose the attacker starts with some belief about that, the prior, and then sees the released output. Pure $\varepsilon$-DP guarantees that the attacker's odds can grow by at most a factor of $e^{\varepsilon}$:

$$\frac{\Pr[\text{Alice is in} \mid \text{output}]}{\Pr[\text{Alice is out} \mid \text{output}]} \le e^{\varepsilon}\,\frac{\Pr[\text{Alice is in}]}{\Pr[\text{Alice is out}]}.$$

Suppose the attacker thinks there is an even chance that Alice is in the data. At $\varepsilon = 1$ the output can push that belief to at most 73%, and at $\varepsilon = 3$ to at most 95%. An attacker who starts at 1% ends at most at 2.7% in the first case and 17% in the second. Those sentences are what a privacy officer can put in a risk assessment. Approximate $(\varepsilon, \delta)$-DP adds a small failure probability $\delta$, and the statement then holds except on outcomes of total probability about $\delta$.

A second statement bounds any test the attacker might run. Write the attack's false-positive rate, the fraction of non-members it wrongly flags, as FPR, and its true-positive rate, the fraction of members it catches, as TPR. Then every membership test against an $(\varepsilon, \delta)$-DP release satisfies

$$\text{TPR} \leq e^{\varepsilon}\,\text{FPR} + \delta.$$

At an FPR of 0.1%, the level at which attacks are usually judged, the bound is a TPR of 0.27% at $\varepsilon = 1$ and 2% at $\varepsilon = 3$. At $\varepsilon = 8$ the bound exceeds 100% and says nothing. That does not mean a model trained at $\varepsilon = 8$ is unprotected. It means the single pair $(\varepsilon, \delta)$ throws information away, and the full privacy curve of the mechanism, which Section 4 uses, gives a much tighter bound.

Bounds describe the worst case, and decision makers also respond to evidence. A privacy audit runs the strongest known attacks against the actual model and reports what they achieved. Earlier audits needed hundreds of training runs. Steinke, Nasr and Jagielski showed how to get a meaningful empirical lower bound on $\varepsilon$ from a single run (Steinke et al., 2023). Jayaraman and Evans documented a large gap between the bounds and what attacks measure in practice (Jayaraman & Evans, 2019). Risk and legal teams can act on a sentence such as "the best attacks we know did no better than guessing". It should be reported next to the bound, though, and never in place of it.

Two more translations help. The first places the release in the organisation's risk register. There, the likelihood of a re-identification incident is multiplied by its impact in fines, notification costs and lost customers, and DP lowers the likelihood in a way that can be defended. Whether a DP release also counts as anonymous data, and so falls outside some data-protection obligations, is a legal question for counsel. The UK Information Commissioner's Office treats DP as a way to produce anonymous statistics but sets no threshold for $\varepsilon$ (summary of ICO guidance). The second translation compares the choice with deployments that peers and regulators have accepted, which Section 8 does.

## 3 The upside: data that DP unlocks

Framed as above, the trade-off is all cost, because the private model is compared with a non-private one. That comparison is often the wrong one. Many projects can only run with a formal guarantee. Legal may not approve the use of the data otherwise, a partner may not share its data, the data may not be allowed to leave a region, or a privacy review may hold the project for months. In those cases the honest comparison is between a private model and no model at all.

The Internal Revenue Service's work with Tumult Labs is a published example. Its older approach suppressed small groups to protect taxpayers, and suppression had climbed above 70% of the cells, which left data of questionable use. A DP release, with its budget tuned to the uses that stakeholders said they needed, replaced it (Tumult Labs). That is a vendor's case study, so its claims are not independently checked, but the pattern is common. So the value of the use case

that DP unlocks belongs on the utility side of the ledger. The cost of privacy then becomes a discount on a revenue line that would otherwise not exist.

## 4 Making the privacy axis comparable

A frontier compares many configurations, possibly built with different techniques, and that only works if every point's privacy is measured the same way. Two values of $\varepsilon$ can describe very different guarantees, and the ways they differ are easy to miss. Table 1 lists what must be held fixed before any two points are compared.

Table 1: What must be the same for every point on a frontier.

| Hold fixed | Why it matters |
|---|---|
| The privacy unit (one record, one user, one user per day) | A guarantee of $\varepsilon = 2$ per record is much weaker than $\varepsilon = 2$ per user when a user has fifty records. |
| The neighbouring relation (adding or removing a record, or replacing one) | For the same mechanism, replacing a record costs roughly twice as much $\varepsilon$ as adding or removing one. |
| The horizon (one release, or a year of monthly retraining) | Repeated releases compose, so the frontier should describe the whole horizon. |
| The value of $\delta$ | Use one value below one over the number of people, for every point. |
| The accountant | Recompute every point with the tightest accountant available, the privacy loss distribution, so that no technique wins only because its paper used a looser bound. |
| The trust model (central, local, shuffled) | The same $\varepsilon$ means different things when a trusted curator adds the noise and when each user adds it. |

Ponomareva and colleagues give a checklist for reporting each of these in their practical guide to DP machine learning, and it doubles as a checklist for normalising a comparison (Ponomareva et al., 2023).

Even with every row of the table fixed, $\varepsilon$ at one value of $\delta$ hides the shape of the guarantee. Two mechanisms with the same $(\varepsilon, \delta)$ can give an attacker very different power. A more robust privacy axis is the strongest attack the full guarantee allows, such as the largest membership TPR at an FPR of 0.1%. Every modern accountant can compute it from the mechanism's privacy curve. For a mechanism that is $\mu$-Gaussian DP, for instance, the bound is

$$\mathrm{TPR} \le \Phi(\Phi^{-1}(\mathrm{FPR}) + \mu),$$

where $\Phi$ is the standard normal distribution function and $\mu$ measures how far apart the attacker's two hypotheses are. Kulynych and colleagues went a step further and calibrated the noise directly

to a target attack risk instead of a target $\varepsilon$. At the same level of risk they got more utility than standard calibration (Kulynych et al., 2024). The attack-facing axis is also the one to show decision makers, with $\varepsilon$ kept as a secondary label.

## 5 Building the Pareto frontier

### 5.1 The candidate space

A point on the frontier is one configuration, made of a technique, its hyperparameters and, when the pipeline has several private steps, a split of the budget between them. The techniques worth comparing for a machine-learning task include DP-SGD from scratch and DP fine-tuning of a public pre-trained model, either the full model, a low-rank adapter or only the last layer. They also include DP-FTRL, which does not need random sampling of the training data, and PATE, which trains teachers on disjoint data and releases their noisy votes. Beyond those come DP synthetic data followed by an ordinary model, output perturbation for convex models, and DP features feeding a non-private model.

Each technique has its own hyperparameters, and most of them move both axes at once. Avent and colleagues point out that DP algorithms have more hyperparameters than their non-private versions for exactly this reason (Avent et al., 2020). For DP-SGD the main ones are the noise multiplier, the clipping norm, the sampling rate, the number of steps and the learning rate.

### 5.2 The procedure

The sweep fixes a grid of target privacy levels and, for each technique and each target, searches for the configuration with the best utility. It works in a fixed order.

1. Pick the targets, for example $\varepsilon \in \{0.5, 1, 2, 4, 8\}$, with the unit, $\delta$ and horizon fixed as in Table 1, and include the non-private model as a ceiling.
2. For each technique, target and hyperparameter setting, let the accountant solve for the noise that meets the target. This step needs no training and takes milliseconds.
3. Train with that noise, evaluate the business metric on held-out data, and repeat with three to five random seeds, because DP training is noisy.
4. Recompute each run's privacy with the common accountant and convert it to the attack-facing axis of Section 4.
5. Keep the runs that no other run beats on both axes.

The last step is the Pareto filter. A configuration is *dominated* when another has no more privacy loss and no less utility, and is strictly better on at least one of the two. The frontier is the set of configurations that are not dominated, and it can be computed with one pass over the points sorted by privacy loss:

```
def non_dominated(points: list[tuple[float, float]]) -> list[int]:
    """Indices of the (privacy loss, utility) points that no other point dominates."""
    order = sorted(range(len(points)), key=lambda i: (points[i][0], -points[i][1]))
    frontier: list[int] = []
```

```
    best = float("-inf")
    for i in order:
        if points[i][1] > best:
            frontier.append(i)
            best = points[i][1]
    return frontier
```

Three practical points decide whether the frontier can be trusted. First, the accountant is free and training is not, so the grid should be pruned before any training starts. For DP-SGD, utility depends mostly on the noise relative to the batch size and on the number of steps, which rules out many settings in advance. Second, the hyperparameters must be tuned again at each target. The best clipping norm, batch size and learning rate change with $\varepsilon$, and a frontier that reuses the non-private settings understates what DP can do. Third, the seeds give each point a confidence interval, and one point should only be said to dominate another when it is clearly better, not just better on average.

When each training run is expensive, Bayesian optimisation can replace the grid. Avent and colleagues map the frontier with multi-objective Bayesian optimisation over measured utility (Avent et al., 2020). A 2025 framework exploits the fact that each frontier point is a maximisation of utility at a fixed privacy level and models the front directly, which needs fewer runs (interactive trade-off framework, 2025).

### 5.3 Paying for the search

Building the frontier is itself an analysis of the private data. Suppose the search tunes on that data and then reports the best run's $\varepsilon$. The reported number is too optimistic, because the choice of the best run leaks information too. Papernot and Steinke showed that this leakage is real, since outliers can shift which settings win. They also showed that it stays modest when each run is private and the number of runs is itself random (Papernot & Steinke, 2022). A follow-up lowers the cost further by tuning on a random subset of the data (NeurIPS 2023 follow-up).

That leaves three honest options. The search can run on public or proxy data from the same distribution, so that only the final training run touches the private data. It can charge its own cost to the budget, using the Papernot–Steinke analysis. Or it can choose a configuration and retrain once, with the report saying plainly that the search itself was not accounted for. For anything shown to a regulator, only the first two will do.

### 5.4 Traps when comparing techniques

A few techniques need care before they are placed on the same plot. PATE's $\varepsilon$ is computed from the actual teacher votes, so it depends on the data, and the number itself leaks unless it is released with its own privacy mechanism. Its points should be marked differently. Public pre-training often moves the whole curve more than any other choice, but it assumes the public data really is public and does not overlap the private data, and that assumption must be stated. DP synthetic data spends its budget once for every later use, so its utility is a frontier per downstream task, and a dataset that is good for one task can be useless for another. Finally, a change of privacy

unit needs new training runs. A guarantee per user needs sampling and clipping per user, and record-level runs cannot be relabelled as user-level ones.

## 6 Fitting the frontier to policy budgets

### 6.1 The policy sets a ceiling, not a weight

A privacy policy can enter the decision in two ways. It can be a weight, so that the team maximises utility minus some price per unit of $\varepsilon$, or it can be a ceiling, so that the team maximises utility below a limit that the policy sets. The ceiling works far better in practice. Nobody can say what a unit of $\varepsilon$ is worth in euros, while a risk owner can say which odds of re-identification they will accept. The ceiling should also be set before anyone sees the utility numbers, because otherwise it tends to drift to wherever the model happens to perform well.

The 2020 census followed this order in reverse, and its example is instructive. The Bureau first fixed accuracy targets drawn from the Department of Justice's redistricting and Voting Rights Act uses. Its engineers then searched for the smallest privacy-loss budget that met them, measuring mean absolute error, percentage error and outliers at every level of geography (Abowd et al., 2022; Census Bureau, 2021). To meet those targets the global budget rose from 4.5 in earlier demonstration data to 12.2 in April 2021 (Census Bureau, 2021), and a policy committee, not the engineers, made the final decision. Whether the constraint is placed on privacy or on utility, the frontier turns it into a single point, and a named owner signs off on that point.

### 6.2 One frontier view per policy

The policy fixes the unit, the horizon, $\delta$ and the ceiling. Many runs can be rescored under a different policy without retraining, because a different $\delta$ or a different horizon only changes the accounting. Only a change of privacy unit or trust model needs new runs. So the experiments can be run once and viewed under each policy, for example a stricter one for the European Union and a looser one for internal use. Each policy draws a vertical line on the plot, and its decision point is the best utility to the left of that line. The distance between two such points is exactly what the stricter policy costs.

### 6.3 Sharing one budget across products

Some organisations set a budget per person across everything they release, for instance "no customer's data may contribute more than a fixed total per year, across all products". Each project then receives a share, and the shares have to be added up correctly. Plain $\varepsilon$ values do not add up tightly under composition, while the parameter $\rho$ of zero-concentrated DP (zCDP) does, so the budget is best allocated in $\rho$. For the Gaussian mechanism with sensitivity $\Delta$, a share $\rho$ buys noise of standard deviation

$$\sigma = \frac{\Delta}{\sqrt{2\rho}},$$

and $k$ releases with shares $\rho_1, \ldots, \rho_k$ cost $\rho_1 + \cdots + \rho_k$ in total.

The allocation itself is an economic problem, and Abowd and Schmutte framed it that way for statistical agencies. Their rule is to set privacy where its marginal cost equals its marginal benefit (Abowd & Schmutte, 2019). Applied to a portfolio of projects, write $U_i(\rho_i)$ for the business value that project $i$ achieves with budget $\rho_i$, which its frontier provides. Maximising the total value within the organisation's budget leads, when each frontier bends the usual way and gains less per unit as the budget grows, to the condition that every funded project gains the same value from its last unit of budget:

$$U_i'(\rho_i) = \lambda \quad \text{for every funded project } i, \qquad \sum_i \rho_i = \rho_{\text{total}}.$$

Here $\lambda$ is the value of one more unit of organisation-wide budget. In practice the rule is greedy. The next unit of budget goes to the project whose frontier is steepest at its current share, until the slopes match or the budget runs out. Systems that manage budgets at this scale exist. Cohere allocates budget to competing requests with weights that let business-critical tasks go first (Cohere, 2023), and Pujol and colleagues treat the budget as a shared resource with an entitled share per analyst (Pujol et al., 2023).

### 6.4 Retraining is part of the budget

A model retrained every month spends its budget twelve times a year, and the frontier should be drawn for that whole horizon. The cadence is then a lever in its own right, because retraining less often leaves more budget for each release. Sage, a DP training platform built at Columbia, handles a stream of data with block composition, so that models can keep training on new data under one global guarantee (Lécuyer et al., 2019). Its "privacy-adaptive training" also follows the constraint-first logic of this section. It grows the data or the privacy budget until a model meets a quality target that the developer set in advance.

## 7 Presenting the decision

The frontier is the engineer's working tool, and the decision maker needs a short menu taken from it. Three to five options, each near a policy ceiling, are usually enough. Table 2 shows the shape of such a menu for a fraud model. Its numbers are illustrative, not measured.

Table 2: An options menu for a fraud model. The numbers are illustrative.

| Option | Privacy statement | Audit result | Business impact | Other effects |
|---|---|---|---|---|
| No DP | No formal guarantee; access controls only | Attack catches 3% of members at 0.1% false positives | Baseline | Legal blocks the partner's data |

| Option | Privacy statement | Audit result | Business impact | Other effects |
|---|---|---|---|---|
| $\varepsilon = 8$ | Weak formal guarantee | No better than guessing | −0.3% fraud caught (−€0.4M a year) | None |
| $\varepsilon = 3$ | An attacker at even odds reaches at most 95% | No better than guessing | −1.1% (−€1.5M a year) | Unlocks the partner's data (+€6M a year) |
| $\varepsilon = 1$ | An attacker at even odds reaches at most 73% | No better than guessing | −4% (−€5.5M a year) | Twice the compute recovers half the loss |

The process around the menu matters as much as the menu. The risk owner, usually the data-protection officer or the risk function, sets the ceiling first. Engineering then fills the menu from the frontier, and the business owner signs off on the remaining utility cost as the named owner of the decision. This mirrors the way security budgets are set, so executives recognise it. The decision, its justification and the audit evidence go into a release record. When the budget runs out or the data changes, the decision can then be reviewed instead of re-argued.

## 8 How others have decided

Published deployments give both a method and a set of reference values. The 2020 census is the most thorough public example of the constraint-first method, as Section 6 described. It also produced a translation that executives understand. A test on 1940 census data compared the noise with sampling. A budget of $\varepsilon = 1$ gave about the error of a random sample of half the population, and $\varepsilon = 4$ about the error of a 90% sample (Petti & Flaxman, 2019).

Industry deployments show the budget machinery at work. LinkedIn's Audience Engagements API gives each advertiser a budget that a central service tracks across data centres, so that nobody can average the noise away by repeating queries. The guarantee is per user, and the service uses the tightest composition bounds available to get more utility from each budget (Rogers et al., 2021). Google has trained more than twenty production language models for its Gboard keyboard with DP-FTRL and public pre-training, at zCDP values of $\rho$ between 0.2 and 2. It now requires a DP guarantee for every new Gboard language model, a policy ceiling applied to a whole product line (Xu et al., 2023).

For reference values, Desfontaines keeps a running list of deployments with their published parameters (Desfontaines). Khavkin and Toch surveyed 140 deployments. They found that commercial and government values of $\varepsilon$ are much looser than those in academic papers, and that they have grown over time in every sector (Khavkin & Toch, 2025). NIST's guidelines for evaluating DP guarantees cover the trade-off and its pitfalls for policy makers as well as engineers (NIST SP 800-226, 2025). Dwork, Kohli and Mulligan interviewed practitioners and found no agreed way to choose $\varepsilon$, which is why they proposed a public registry of deployments (Dwork et al., 2019).

Reference values are a sanity check, not a justification. A value is defensible because the menu, the audit and the policy support it, and the fact that a large company chose something similar adds comfort but no argument of its own. One gap remains in the literature. No source found for this note converts $\varepsilon$ into a monetary risk figure, so that step stays with each organisation's own risk register, guided by the economic framing of Abowd and Schmutte.

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